Preparation Memory and Energy-Ordering Reversals in Exact Unforced Flows

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Analytic and computer-assisted results; specialist novelty review remains open.
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Abstract

We study phase-dependent energy ordering in two explicit, globally smooth three-component solutions of the unforced Navier–Stokes equation on the torus. The solutions are independent of one spatial coordinate: a decaying transverse shear transports a vertical velocity field. Their Fourier representation is an infinite dissipative nearest-neighbour chain. Initially identical mode-energy summaries do not determine subsequent total energy. An exact cosine law reduces the phase contrast to a two-column Gram entry. We give an explicit all-time no-reversal estimate for family (i) throughout a nonzero coupling interval, prove that normalized late amplitudes are entire in coupling, and derive local stability and parity rules for the number of ordering reversals. Two computer-assisted examples exhibit one reversal each, in opposite directions; their certificates include a uniform near-zero truncation bound and are kept separate from the analytic results. Both preparation energies decrease throughout. The flows are unforced; no laboratory validation is involved.

1 Question, baseline and contribution

A dissipating system can retain information not contained in its initial energy. The relevant fluid baseline is non-normal shear dynamics: decaying eigenmodes alone need not determine transient response [6]. Our energy-ordering contrast does not require either preparation's total energy to grow. Relaxation-ordering phenomena, including the Markovian Mpemba mechanism, offer another comparison [7]. Here the question is narrower: for an exact unforced fluid family, can one determine the entire sign history of the difference between two preparation energies? Exact decaying bar states and enhanced dissipation for related torus flows are established baselines [8, 9]. Those results make clear why the reduction itself is not the proposed novelty. The object here is the sign history of a cross-preparation energy contrast, with both energies decreasing, rather than a new shear solution or damping mechanism. The classical lineage includes Kelvin's sheared disturbances and Orr's stability analysis [1, 2]. Shear-enhanced scalar mixing and vortex relaxation are also established [3, 4]. The reduction belongs to the standard two-and-a-half-dimensional class [5]. In particular, the result is not Orr transient amplification: neither preparation's total energy grows; the signed *difference between* their energies changes sign.

The contribution proposed for this manuscript is the explicit family and its accounted infinite-dimensional sign analysis. Shear reductions, passive advection–diffusion, phase gauges, Dyson

expansions, analytic perturbation and implicit-function arguments are standard tools. Specialist review of closely related transient-energy and observability results remains outstanding; this preprint does not claim completed external peer review or established priority.

2 An exact unforced reduction

On the 2π-periodic three-torus let spatial brackets denote normalized average, and set $\nu = 1/16$. For integer horizontal vectors k, q_0 and a horizontal vector a satisfying

$$a \cdot k = 0, \qquad a \cdot q_0 = 2c,$$

define

$$v_\theta(x,t) = a\mathrm{e}^{-\rho t}\cos(k \cdot x + \theta), \quad \rho = \nu|k|^2, \qquad u_\theta = v_\theta + w_\theta e_z,$$

where all fields are independent of z and

$$\partial_t w_\theta + v_\theta \cdot \nabla w_\theta = \nu \Delta w_\theta, \qquad w_\theta(x,0) = \sin(q_0 \cdot x) + \cos((q_0 + k) \cdot x).$$

The horizontal self-advection vanishes. All terms involving a z derivative vanish as well, and the remaining convection has zero divergence. A spatially constant pressure therefore makes u_θ an exact unforced incompressible Navier–Stokes solution. Positive diffusion and the smooth prescribed shear give global smooth evolution for this subclass.

We use the following two realizations:

Family	k	q_0	a	ρ
(i)	$(8,-1,0)$	$(0,2,0)$	$(c/8,c,0)$	$65/16$
(ii)	$(1,1,0)$	$(0,-2,0)$	$(c,-c,0)$	$1/8$

The second is a coordinate realization of the chain used in the computer-assisted examples of Section 6. For $q_m = q_0 + mk$, the chains q_m and $-q_m$ are disjoint. Writing the positive-chain coefficient as b_m gives

$$b_m' = -\lambda_m b_m - \mathrm{i}c\mathrm{e}^{-\rho t}(\mathrm{e}^{\mathrm{i}\theta} b_{m-1} + \mathrm{e}^{-\mathrm{i}\theta} b_{m+1}), \qquad b(0) = -\frac{\mathrm{i}}{2}e_0 + \frac{1}{2}e_1.$$

Here $\lambda_m = \nu|q_m|^2$:

$$\lambda_m^{(i)} = \frac{65m^2 - 4m + 4}{16}, \qquad \lambda_m^{(ii)} = \frac{(m-1)^2 + 1}{8}.$$

The vertical energy is $\frac{1}{2}\langle w^2\rangle = \sum_m |b_m|^2$; there is no additional factor two. The phase-independent horizontal energy is $|a|^2\mathrm{e}^{-2\rho t}/4$.

3 Phase contrast and monotone individual energies

On real $\ell^2(\mathbb{Z})$ put $(Jr)_m = r_{m-1} - r_{m+1}$. Then $J^* = -J$ and $\|J\| \le 2$. Let

$$r_j' = -\Lambda r_j + c\mathrm{e}^{-\rho t} J r_j, \qquad r_j(0) = e_j, \quad j = 0, 1. \tag{1}$$

A diagonal phase gauge followed by multiplication by $(-\mathrm{i})^m$ converts the Fourier evolution to this real equation.

Proposition 3.1 (Exact cosine law)**.** *With* $C_c(t) = \langle r_0(t), r_1(t)\rangle$,

$$E_\theta(t) = A_c(t) - \frac{1}{2}C_c(t)\cos\theta, \quad A_c = E_v + \frac{1}{4}(\|r_0\|^2 + \|r_1\|^2). \tag{2}$$

In particular $E_\pi - E_0 = C_c$ *and*

$$C_c'(t) = -2\langle r_0, \Lambda r_1\rangle, \qquad C_c(t) = c(\lambda_0 - \lambda_1)t^2 + O(t^3).$$

Proof. The transformed initial vector is $\frac{\mathrm{i}}{2}(-e_0 + \mathrm{e}^{-\mathrm{i}\theta}e_1)$. Expanding its evolved squared norm gives (2). In differentiating the real inner product the two J terms cancel. At zero, $C = C' = 0$ and $C'' = 2c(\lambda_0 - \lambda_1)$. Finite-support initial vectors justify the derivatives; a convergent time-Taylor series at zero is not assumed. □

For any real or complex column combination r, $(\|r\|^2)' = -2\langle r, \Lambda r\rangle \le 0$ (with the Hermitian inner product for complex combinations). Adding the decreasing horizontal energy proves that both preparation energies decrease. Ordering reversal is not energy growth. The initial per-mode Hermitian covariances are independent of θ, and the corresponding pure Stokes energies do not separate.

Remark 3.2 (Prediction from the initial covariances)**.** Fix the family, coupling and prediction time. A deterministic energy predictor given only the initial per-mode Hermitian covariances must return the same value z for preparations $\theta = 0$ and $\theta = \pi$. The triangle inequality and (2) imply

$$\max\{|z - E_0|, |z - E_\pi|\} \ge \frac{|C_c|}{2}, \qquad \frac{(z - E_0)^2 + (z - E_\pi)^2}{2} = (z - A_c)^2 + \frac{C_c^2}{4}.$$

The midpoint $z = A_c$ attains both two-preparation optima; the second expression is the squared risk for equal prior probabilities on these two preparations. This is an elementary consequence of their identical observations, not a limitation on predictors supplied with distinguishing phase-sensitive or later-time data. At a crossing $C_c = 0$ this particular prediction floor vanishes, so a crossing need not be the time of greatest prediction ambiguity.

4 A uniform all-time no-reversal band

Theorem 4.1. *For family (i), every* $0 < c \le 1/3$ *and every* $t > 0$ *satisfy*

$$C_c(t) \le -c\mathrm{e}^{-t/2}\min\left\{\frac{37}{1152}t^2, \frac{35422319}{10201651200}\right\} < 0.$$

Proof. Let $\Gamma = 2c(1 - \mathrm{e}^{-\rho t})/\rho$. Contractivity of the heat semigroup and the ordered integration simplex bound the nth Dyson term by $\mathrm{e}^{-t/4}\Gamma^n/n!$ for a unit column. Only odd total orders survive the phase-flip contrast. Its first term is

$$C_1 = -c\left[\mathrm{e}^{-t/2}F_{63/8}(t) - \mathrm{e}^{-65t/8}F_{1/4}(t)\right], \quad F_a(t) = \frac{1 - \mathrm{e}^{-at}}{a}.$$

Thus

$$|C - C_1| \le \mathrm{e}^{-t/2}\{\sinh(2\Gamma) - 2\Gamma\} \le \mathrm{e}^{-t/2}R, \quad R = \mathrm{e}^{2\Gamma}(2\Gamma)^3/6.$$

Set $\delta = 61/16$ and $L_1 = -\mathrm{e}^{t/2}C_1$. An integral representation gives $L_1' > 0$ and $L_1 \ge c\delta t^2\mathrm{e}^{-(63/8)t}$. For $0 < t \le 1/4$, $\mathrm{e}^2 < 8$, $2\Gamma \le 4ct$, and $\mathrm{e}^x \le (1 - x)^{-1}$ give

$$\frac{L_1}{ct^2} \ge \frac{61}{128}, \qquad \frac{R}{ct^2} \le \frac{(32/3)c^2t}{1 - 4ct} \le \frac{4}{9}.$$

The difference is $37/1152$. For $t \geq 1/4$, monotonicity gives $L_1/c \geq 61/2048$ while

$$R/c \leq \frac{(64/65)^3}{54(1-64/195)}.$$

Subtracting yields the stated positive rational constant. □

At $c = 0$ the contrast vanishes identically; negative c reverses its sign. The failure of this sufficient estimate at a larger coupling is not itself a reversal or a physical phase transition.

5 Late amplitudes and reversal-count structure

Let $\lambda_* = \min_m \lambda_m$ and s be its unique minimizing index. These are $(1/4, 0)$ in (i) and $(1/8, 1)$ in (ii). Set $z_j = \mathrm{e}^{\lambda_* t} r_j$ and $H = \Lambda - \lambda_*$.

Theorem 5.1 (Analytic late contrast)**.** *There are entire functions $\alpha_j(c)$ such that $z_j(t,c) \to \alpha_j(c) e_s$ locally uniformly in complex c. For real c,*

$$\mathrm{e}^{2\lambda_* t} C_c(t) \longrightarrow L(c) = \alpha_0(c)\alpha_1(c), \quad |\alpha_j^{(n)}(c)| \leq (2/\rho)^n.$$

The entire function L is odd and nonzero, with $L'(0) = -8/63$ in (i) and $L'(0) = 4$ in (ii). Its zeros are isolated and finite on each compact real interval.

Proof. On $|c| \leq R$ the bounded-perturbation series is uniformly dominated by $\sum_n (2R/\rho)^n/n!$. Hence z_j is entire and bounded by $\mathrm{e}^{2R/\rho}$ uniformly in time. Let P project onto e_s and let $Q = I - P$. The positive gap δ of H on Q gives

$$\|Q z_j(t)\| \leq \mathrm{e}^{-\delta t} + 2R\mathrm{e}^{2R/\rho} \int_0^t \mathrm{e}^{-\delta(t-u)} \mathrm{e}^{-\rho u}\, \mathrm{d}u \to 0$$

uniformly on the disk. Also $\|(P z_j)'\| \leq 2R\mathrm{e}^{2R/\rho}\mathrm{e}^{-\rho t}$ is integrable. The resulting uniform holomorphic limit is entire. At real c, the normalized propagator is contractive. Differentiating the equation and integrating its forcing inductively gives $\|\partial_c^n z_j\| \leq [2(1-\mathrm{e}^{-\rho t})/\rho]^n$. Pass to the limit.

For $De_m = (-1)^m e_m$, $DHD = H$ and $DJD = -J$, so $\alpha_j(-c) = (-1)^{s+j}\alpha_j(c)$. The first Dyson term gives the two stated derivatives of L. The identity theorem now excludes accumulating zeros. For complex c, L is the bilinear product of amplitudes, not an antiholomorphic squared norm. □

Proposition 5.2 (Local stability and parity)**.** *Fix $c_0 > 0$ with $L(c_0) \neq 0$, and assume every positive-time zero of C_{c_0} is simple. Their number is finite and locally constant in c. If σ_0 and σ_∞ are the early and late signs, the number n of simple crossings satisfies $(-1)^n = \sigma_0\sigma_\infty$.*

Proof. The nonzero quadratic early coefficient excludes a common interval next to zero. Uniform convergence to nonzero L excludes a common late interval. Simple zeros on the remaining compact time interval cannot accumulate: an accumulation would have both $C = 0$ and $C_t = 0$. The implicit function theorem continues each root, and a positive off-root compact minimum excludes new ones nearby. Each crossing flips sign. □

Count changes therefore require a finite-time multiple zero or a late-amplitude cancellation, under the stated hypotheses. This is a necessary event list, not a global count classification or a claim that every event changes count.

6 Two certificate-supported reversal examples

The computations and their analytic supplement [10] support:

Family, coupling	Early sign of C	Unique zero bracket	Late sign
(ii), $c = 1/4$	$+$	$[2199/128, 10997/640]$	$-$
(i), $c = 8$	$-$	$[1307/2400, 109/200]$	$+$

These brackets are nondimensional.

6.1 Finite-chain error charged to the continuum

For a chain truncated on $[-M, M]$, $M = 24$, let v_j be extended by zero. With $a_t = \mathrm{e}^{-\lambda_* t}$ and the above Γ, set

$$\beta = a_t \Gamma^M \mathrm{e}^{\Gamma}/M!, \qquad \eta = 2\beta.$$

Dyson support implies agreement before order M, so $\|r_j - v_j\| \le \eta$ and $|C - C_M| \le 2a_t\eta$. If $p(k) = \max_{|m|\le k} \lambda_m$, then

$$p_i(k) = 65k^2/16 + k/4 + 1/4, \quad p_{ii}(k) = k^2/8 + k/4 + 1/4.$$

Expanding these polynomials in falling factorials bounds the weighted tail by $\|\Lambda(r_j - v_j)\| \le p(M+1)\eta$. Consequently

$$|\,\langle r_0, \Lambda r_1\rangle - \langle v_0, \Lambda v_1\rangle\,| \le 2p(M)a_t\eta + p(M+1)\eta^2.$$

Near zero, a uniform charge is necessary:

$$\frac{|C - C_M|}{t^2} \le \frac{4(2c)^{24}\tau^{22}}{24!(1 - 2c\tau)} \quad (0 < t \le \tau < 1/(2c)).$$

The finite-chain certificates control C_M; this bound charges the whole-chain difference uniformly down to $t = 0$. Safe bounds are $C \ge (3/100)t^2$ up to $3/40$ in (ii) and $-C \ge (299/10)t^2$ up to $1/300$ in (i).

6.2 Evidence boundary and reproducibility

The accompanying supplement provides interval sources, comparison inputs, a replay driver, and the analytic continuum-error and late-time bounds. The driver starts from unit initial columns and recomputes both finite-chain trajectories, the zero-window derivative certificates and all 4,224 early subcells. It checks the sign runs, straddle cells and adjoining interval endpoints. The zero-window calculation uses the trajectory computed during the same invocation. These are same-code reproductions, not independent ODE implementations.

The supplement distinguishes interval predicates from rounded display endpoints and requires the corrected whole-chain coefficient. Review of the Cauchy and Dyson bounds is part of checking the continuum result; matching a sampled trajectory alone is insufficient. Analytic Theorems 4.1–5.1 do not depend on the finite calculations.

7 Prospective use and limitations

A test of the prediction must fix its preparation, coupling, observation times, normalization and instrument error before measurement. It must compare against the same ordinary unforced equations and retain numerical and measurement errors separately. The cosine law predicts a relational quantity; a single declining energy record does not reveal the phase.

A finite error budget can hide the contrast early and late while exposing it at an intermediate time, although the underlying coupling is continuous. The equality $E_\theta = E_{-\theta}$ has a rigid-motion explanation in the specified preparation. Open questions include the global reversal-count diagram, optimal no-reversal thresholds, and robustness to physical preparation and readout errors.

Drafting and verification disclosure

Luqman Mahmood directed the research and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted mathematical work, computation, checking and editing. Statements have different evidence classes: written conditional proofs, computer-assisted certificates, numerical demonstrations, and prospective protocols. These are labelled as such in the text and do not constitute external peer review. No claim of literature priority follows from computational verification. The paper and mathematical data are licensed CC BY 4.0. Software source and executable verification scripts are excluded from that grant, with rights reserved unless a file explicitly states otherwise.

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